User Manual for Madhacker BMSOne

Introduction

The Madhacker BMSOne is a universal Battery Management System (BMS) designed for 2S-6S lithium battery packs, compatible with ArduPilot vehicles release 4.4.0 and later. It provides real-time monitoring of battery parameters including total voltage, individual cell voltages, current, state of charge (SoC), and temperature. The device supports customizable configurations for different battery chemistries (e.g., LiPo, Li-ion, other) and capacities, with features like automatic cell count detection, voltage-based or capacity-based SoC estimation, and optional thermistor-based temperature sensing.

Key benefits include:

- User-configurable voltage and percentage tables for accurate SoC calculation.
- Multiple configuration banks for storing and switching settings.
- Calibration options for voltage and current sensors to improve accuracy.
- Plug-and-play compatibility with 3DR Solo.
- Firmware upgradable via serial interface.
- Reverse polarity protection

Note: This manual is based on firmware versions up to 2.5 and hardware revisions supporting 2S-6S. Always refer to madhacker.org for the latest updates. Calibration and advanced configurations are optional but recommended for optimal performance.

Hardware Specifications

- **Dimensions:** Less than 50 × 40 × 9 mm (balancing connector is the tallest component)
- **Weight:** Approximately 7.9 g
- **Connectors:**
    - Balancing connector (varies by cell count: 3S, 4S, 5S, or 6S)
    - USB-C for configuration and firmware updates (115200 baud, 8N1)
    - Battery input/output (B+ / B−)
    - Optional temperature sensor input (for 100K NTC thermistor)

- **Power for Configuration:** Connect via USB-C. For proper calibration, ensure the battery is connected. (If the BMSOne is an older version that connects via FTDI cable, disconnect +5V from the serial adapter to avoid conflicts.)
- **Mounting:** Includes mounting holes suitable for integration into devices like the 3DR Solo GPS bay.

The hardware supports auto-detection of cell count, and the same firmware works across 2S-6S variants (differentiated only by the balancing connector).

Setup and Basic Operation

1. **Connection**
    - Connect the battery pack to the B+ / B− terminals.
    - Attach the balancing wires to the balancing connector in the correct order (lowest to highest cell).
    - Connect the Load+ / Load- terminals to the device being powered.
    - For temperature monitoring, solder a 100K NTC thermistor to the designated pins (if not using internal sensor).
    - Connect to a host system (e.g., ArduPilot: BATT_MONITOR=5) via the SMBUS/I2C-2 interface. Connect SCL and SDA only, leaving the GND pin disconnected.
2. **Initial Boot**
    - The device auto-detects cell count (default: SY mode).
    - It reads configuration from the current bank (SR command executes automatically).
    - Default SoC mode is voltage-based (S0), which is recommended for most users.

Configuring the BMSOne

- Use a serial terminal (e.g., Tera Term) connected via USB-C at 115200 baud.
- Send commands followed by Enter. Changes are temporary until written with SW.
- All commands are sent via the serial interface.

- Commands are case-sensitive and do not require parameters unless specified.

Commands (case sensitive)

- **S0** — Select voltage-based SoC (recommended mode; estimates remaining capacity from voltage table)
- **S1** — Select capacity-based SoC (tracks mAh consumed; assumes full charge at boot)
- **S!** — Disable thermistor; report fake 22.5°C temperature
- **SC** — Calibrate cell voltages (e.g. SC3740,3746,3752,3749,3751 for a 5S pack; see Calibration section)
- **SD** — Show internal data (for debugging)
- **SH** — Enable 100K thermistor for actual temperature reporting
- **SI** — Calibrate current sensor (e.g. SI10220 for 10.220A; see Calibration section)
- **SM** — Set total battery capacity in mAh (e.g. SM10000 for 10Ah; optional in S0 mode)
- **SP** — Set percentage table (e.g. SP0,8,21,27,41,49,57,64,72,80,85,88,98,99,100; see Tables section)
- **SR** — Read config from current bank (automatic on boot)
- **SR1** — Read config from Bank 1
- **SR2** — Read config from Bank 2
- **SR3** — Read config from Bank 3
- **SS** — Show current configuration
- **SW** — Write configuration to current bank (makes changes permanent)
- **SW1** — Write to Bank 1
- **SW2** — Write to Bank 2
- **SW3** — Write to Bank 3
- **SW!** — Write to all banks
- **SV** — Set voltage table (e.g. SV3500,3525,…; see Tables section; values in mV per cell for new firmware, or total pack voltage for old)
- **SO** — Force old protocol (4S only; spoofs 4S for compatibility with older ArduPilot)

- **SN** — Enable new protocol (2S-6S support)
- **SY** — Enable auto cell count detection (default)
- **ST** — Invert current multiplier (for reversing current direction sensing)
- **SJ** — Set current multiplier (e.g. SJ-7.5 to adjust scaling; use positive/negative as needed)

Important Notes:

- Bank is selected by pressing the physical button on the BMSOne. Each press changes to the next bank, cycling 3 banks (1→2→3→1→2→3).
- Default Bank is the most recent Bank written. To change the default Bank issue SW1, SW2, or SW3 (e.g. SW2 makes Bank 2 the default on reboot).
- Use multiple banks (1–3) to store different configurations (e.g. 1 for LiPo 5200mAh, 2 for Li-ion 6000mAh, 3 for Li-ion 8000mAh).
- After changes, use SW to save to the current Bank; reboot to revert unsaved changes.

Calibration Procedures

Calibration improves accuracy but is not mandatory (devices are pre-calibrated). Perform with battery connected and a precise multimeter or current source. Refer to madhacker.org for additional details.

Voltage Calibration (SC)

- Measure each cell's voltage with a multimeter.
- Enter values in millivolts (mV), comma-separated, for all cells.
- Example for 5S pack: SC3740,3746,3752,3749,3751 (corresponds to 3.740 V, 3.746 V, etc.)
- For partial calibration, omit values (e.g. SC,,3300, for cell 3 only)
- Save with SW after calibration.

Current Sensor Calibration (SI)

- Pass a known current through the device (e.g. 10 A; higher currents yield better accuracy).
- Measure the actual current.
- Enter value in mA (e.g. SI10220 for 10.220 A).

- If direction is wrong, use ST to invert.
- Adjust multiplier if needed: SJ-7.5 (example for scaling).

Tips: Calibrate voltage first. Use a stable power source. Save changes with SW.

Voltage and Percentage Tables

Tables linearize SoC estimation. Use SV for voltage points (in mV per cell for new firmware; total V for old) and SP for corresponding percentages. There are 15 points per table.

New Firmware Tables (BMSOne firmware 2.0 and up uses cell voltage)

LiPo (per cell)

- Voltage: SV3500,3525,3550,3575,3600,3625,3675,3700,3725,3750,3800,3825,3950,4000,4200
- Percentage: SP0,8,21,27,41,49,57,64,72,80,85,88,98,99,100

Li-ion (per cell)

- Voltage: SV2800,2900,3000,3100,3200,3320,3400,3500,3600,3700,3800,3860,3900,3960,4200
- Percentage: SP0,7.3,14.6,21.9,29.2,36.5,43.8,51.1,58.4,65.7,73.0,80.3,87.6,94.9,100

Old Firmware Tables (BMSOne firmware below 2.0 uses pack voltage)

4S LiPo (total pack V)

- Voltage: SV14.0,14.1,14.2,14.3,14.4,14.5,14.7,14.8,14.9,15.0,15.2,15.3,15.8,16.0,16.8
- Percentage: SP0,8,21,27,41,49,57,64,72,80,85,88,98,99,100

4S Li-ion (total pack V)

- Voltage: SV10.4,11.0,11.3,11.6,11.9,12.2,12.6,12.9,13.2,13.5,13.8,14.2,14.5,14.8,16.8
- Percentage: SP0,4,12,20,29,37,45,53,62,70,78,86,95,99,100

5S Li-ion (total pack V)

- Voltage:
  SV14,14.5,15.0,15.5,16.0,16.6,17.0,17.5,18.0,18.5,19.0,19.5,20.0,20.5,21.0
- Percentage:
  SP0,7.3,14.6,21.9,29.2,36.5,43.8,51.1,58.4,65.7,73.0,80.3,87.6,94.9,100

5S LiPo (total pack V)

- Voltage:
  SV17.5,17.6,17.7,17.8,18.0,18.1,18.3,18.5,18.6,18.7,19.0,19.1,19.75,20.0,21.0
- Percentage:
  SP0,7.3,14.6,21.9,29.2,36.5,43.8,51.1,58.4,65.7,73.0,80.3,87.6,94.9,100

6S Li-ion (total pack V)

- Voltage:
  SV18.0,18.2,18.5,18.8,19.2,19.6,20.0,20.4,20.8,21.2,21.6,22.0,22.4,22.8,25.2
- Percentage:
  SP0,7.3,14.6,21.9,29.2,36.5,43.8,51.1,58.4,65.7,73.0,80.3,87.6,94.9,100

Usage: Send the full command (e.g. SV followed by comma-separated values). Use S0 mode for voltage-based SoC. Save with SW.

Updating Firmware

Firmware updates require avrdude (available for Windows, Linux, Mac) and a serial connection.

1. Open a serial terminal (e.g. Tera Term) on the device's COM port (e.g. COM9) at **1200 baud**.
2. Close the terminal immediately to enter bootloader mode.
3. Run the avrdude command in Command Prompt/Terminal to flash the HEX file.

Example commands (replace COM10 with your port and HEX file as needed):

text

avrdude -patmega32u4 -cavr109 -P COM10 -b115200 -D -Uflash:w:BMSOne2.4.hex:i

Alternative (to open bootloader):

text

avrdude -c avr109 -p m32u4 -P COM4 -b 1200

For USBASP flashing:

text

avrdude -patmega32u4 -cusbasp -D -Uflash:w:BMSOne2.5.fix.hex:i

Warnings: Ensure correct port and baud. Backup configurations before updating. Download HEX files from madhacker.org.

Advanced 3DR Solo Configuration (Warning Thresholds)

For 3DR Solo running Open Solo 4 the cell voltage warning can be adjusted. Edit constants in /usr/bin/shotManagerConstants.py:

- BATT_CAPACITY_WARN = 500 # Alert if full charge capacity degrades to this (mAh)
- BATT_CELL_LOW_WARN = 2650 # Low cell voltage warning (mV)
- BATT_CELL_DIFF_WARN = 300 # Cell voltage differential warning (mV)

Restart the software after changes.

Troubleshooting

- **No Response** — Check serial baud rate and port. Reboot device.
- **Inaccurate SoC** — Verify tables match battery type; recalibrate.
- **Compatibility Issues** — Use SO for old 4S-only systems.
- **Temperature Errors** — Toggle between S! and SH.

For more support, visit madhacker.org or check community forums.

Safety Notes

- Always follow battery manufacturer’s safety recommendations.
- Monitor for heat during operation.
- Do not short-circuit or over-discharge.
- Calibration requires precision tools; improper settings can damage batteries.